
Powering West Africa: modelling sustainable long-term electricity planning

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Abstract: This paper presents a tree-like exploration method to analyse strategies for the development of grids of the future in West Africa. The approach helps generate plausible future grids, based on demand requirements and resource availability, by either building power plants or transmission lines. These grids are generated via a modelling framework, *Spark!*, which captures key components of the power system, as well as relationships between them. *Spark!* analyses financial, environmental and reliability performances, and also addresses concerns caused by high renewable penetration in large-scale power grids. Contrary to other models used to address long-term planning in African countries, *Spark!* performs long-term planning of energy resource use to satisfy the demands, but also simulates grid operations in real-time. It enables decision makers to examine long-term effects of decisions made, but also observe how the system responds to these decisions.

Keywords: energy; system modelling; electricity; grid simulation; power grid; African power sector; energy planning; sustainable development; energy planning model; Africa; West Africa.

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1 Introduction

One of the major challenges faced by Africa's power sector is that grid systems in various African countries do not function very well, presenting an electricity access rate of just over 40%, and leaving over 600 million people without access to electricity (International Energy Agency, 2017). Such a low rate negatively affects the African population, specifically with pernicious effects on health, education, business growth and security (Oyuke et al., 2016). The weakness of the power sector has considerably constrained the economic growth and development of the African continent, which has struggled to sustain GDP growth (Castellano et al., 2015). The continent recorded an average of 690 hours of power outages in 2015 (World Bank, 2017), which approximately cost Africa 2–5% of its annual GDP (Africa Progress Panel, 2017). The limited access to electricity, the lagging power transmission network and the inadequate generation systems hamper economic growth, and limit employment creation and investment. Those statistics provide insight for the whole continent, though it is interesting to note disparities between populations, and also regions and countries. According to a survey conducted by Eaglestone Securities (2016) over 36 countries, 94% of urban population have access to the electric grid, whereas only 45% do in rural areas. Across nations, many North African and island countries achieve high rates of access, as opposed to some sub-Saharan countries which extend electricity to only a third or less of the country, namely Burundi (17%), Burkina Faso (25%), Sierra Leone (29%), Niger (30%), Guinea (31%), Liberia (31%) and Mali (32%). Still according to the same survey, Western and Eastern regions lag behind other regions in expanding the grid. In a report released by the Nigerian Ministry of Budget and National Planning, out of 12.5 GW of installed capacity in Nigeria, only 3.6 GW is operational for generation (MBNP, 2017). Losses in transmission and distribution in this country are as high as 50%, which put the actual power capacity distributed at around 1.9 GW (MBNP, 2017). Almost 85% of the installed capacity is unused, which is an obvious sign of lack of effectiveness in grid performance, especially, for a country as economically powerful and populous as Nigeria.

If the issue of effective functioning of the grid is not resolved, the long-term sustainability and financial viability of the power sector in Africa will undoubtedly suffer. Africa presents a great diversity in energy resources (Althusmann and Ruppel, 2016), which, if harnessed, would enable the continent to stay clear from over-reliance on a single resource, and constitute a great enabler for technology development. Africa is a land of renewable energy opportunities, with a vast unexploited resource pool, that can supply power for all, and at modest cost. Truth be told: Africa's energy sector is resourceful, yet not powerful. Ameliorate electricity supply, distribution and transmission systems, in order to better energy access and ultimately offer avenues for economic growth, are challenges of top priority. Today, African leaders recognise the urge and necessity for a reliable and effective electricity supply and affordable distribution network. Governments have combined efforts, through partnerships with developed nations, to take actions in this sense. One of the most recent and pertinent initiative is the African Development Bank's *New Deal on Energy for Africa*, launched in 2015, aimed at helping achieve continent-wide electricity access by 2025. The focus is placed on increasing the on-grid generation, on-grid transmission and grid connections, off-grid generation as well as access to clean cooking energy (AfDB, 2017). Regional power pools have been established as a means to minimise the costs of market fragmentation for

countries with limited economic, political or geographical coherence, to develop ambitious strategies for sustainable resource mix across countries and to help them benefit from a better interconnection (WEC, 2005).

Effective resource planning is no easy task. The difficulty that arises is the ability to accurately anticipate changes that might occur in the future. Those changes may take place at various levels, namely fuel prices, electricity demands, technology evolution, regulations, etc. The use of energy models is of great help. They have a documented history of being used to address energy system issues, and to serve as decision support system, given their ability to provide insights into how power systems function and may evolve in years to come. Such analytical efforts hold great potential, especially in Africa. For instance, the West African Power Pool (WAPP) used a modelling platform, named MESSAGE (IRENA, 2015b). This energy model was used as a planning tool, to help study investment alternatives via various scenarios, and assess the economic, environmental and social implications of a given investment path.

Caution, however, needs to be observed, when it comes to the use of energy models. Although energy models are mainly used to answer energy planning questions, these types of models are not appropriate for power grid simulation. Energy system models capture the behaviour of the entire energy system, its impact on the wider economy, over a medium and long-term horizon. By 'entire energy system', understand the combination of the electricity sector, the heat, gas, mobility, and other sectors as appropriate (Abrell and Weigt, 2012). In these models, the power sector is endogenous, and controlled by the combined behaviours of supply sectors and end-use sectors driven by exogenous demands (Deane et al., 2012). The electricity sector is thus coarsely represented, with no regard for reliability, flexibility, unit commitment and dispatch. These limitations hamper their ability to provide trusted results, regarding power system planning. By contrast, power system models focus only on the electrical power system (supply, transport and use of electricity). Electricity (or power) is a specific form of energy, and is the main form we are interested in. The difference between energy and power system models resides in their focus or application, and the questions they are answering. Energy systems model tend to cover a much broader range of problems, for instance, linkages between energy and the economy (Messner and Schrattenholzer, 2000; Nakata et al., 2011), or optimal long-term energy supply strategy based on cost minimisation (Hainoun et al., 2010), among others. Power systems are literally and exclusively focused on electricity supply, demand and transmission.

A better way to address issues related to power system planning, short-, and long-term would be to use actual power system models. They represent the power sector in a more detailed manner, and have more propensity to capture phenomenon critical to that sector. Such models should enable long-term projections validation via system operations in the short-term. In other words, the viability of the suggested projections made over years is tested by how the power grid system responds to them, on a daily basis. Otherwise, it would not be possible to reconcile long-term planning with short-term grid operations. Only this way can decision makers appreciate the appropriateness of projections and long-term policies. Corrective actions may thus be taken if necessary.

This study develops an exploration method to quantitatively analyse the current status of power generation and predict the composition of the future generation profile, and the associated costs, reliability and gas emissions, in West Africa. This method includes the

deployment of renewable sources and supply side energy-efficiency policies, in order to balance the grid system. The modelling platform used, *Spark!*, performs long-range electricity planning, as well as real-time grid dispatch operations. Contrary to the existing models in the literature, the model presents a more detailed representation of the grid, with weather and technology-specific information as input, to account for large-scale high renewable penetration grids.

The remainder of the paper is as follows: Section 2 presents the modelling approach. Application is described in Section 3. Section 4 presents the simulation results and Section 5 concludes.

2 Modelling approach

2.1 Existing power system models

Power system models can be grouped into three: power flow models (Milano, 2010; Preece, 2013), capacity expansion models (CEM) like RPM (Mai et al., 2013), ReEDS (Short et al., 2011), or SWITCH (Fripp, 2012), and production costs models (PCM), like PLEXOS (CAISO, 2010), PROMOD (Lohmann, 2015) or WILMAR Planning Tool (Larsen, 2006).

Power flow models are mainly used for power-flow study. They address issues related to various aspects of alternative current (AC) power parameters, namely voltages, voltage and phase angles, real power and reactive power, which are not within the scope of this paper. Also, running long-term simulations on these models, for large-scale grid would require heavy computational capabilities, and very long wall clock time.

By power system, we mean network of entities interacting with one another to enable supply, transport, distribution and use of electricity. In that sense, the models we are interested in would represent elements or components able to perform these activities. Unlike power flow models, our models of interest, PCM and CEM, would rank at the activity level, with the focus on activity development to implement strategies. In our case, these strategies are plans or actions in an attempt to ensure reliability, sustainability and profitability of the system.

PCM simulate routine power system operations on shorter time scales (mostly hourly) using detailed load, transmission, storage and generation fleet data. They help answer security-constrained unit commitment and security-constrained economic dispatch questions, and perform transmission analysis and planning, considering economic, environmental and operational constraints (Stoll et al., 2016). Typical outputs include location marginal prices (LMP), emission rates, generating unit usage, transmission rate usage, etc. CEM on the other hand are built to answer long-term planning questions. They provide a high-level (yearly or multi-year basis) view of the evolution of power systems. Typical outputs include generation and transmission capacity expansions or retirements, annual generation, etc. Several authors have resorted to CEM for long-term electricity supply-demand studies (Gnansounou et al., 2007; IRENA, 2013a, 2015a; McKinsey&Company, 2015; Ouedraogo, 2017; Panos et al., 2016; Eberhard et al., 2011; Rosnes and Vennemo, 2012).

With little doubt, CEM are the preferred type of model to analyse possible future paths for power systems, identify most efficient and environmentally friendly way for power systems development and assess feasibility and consequences of policies taken along the way. However, because of their structure and granularity level, they do not represent operational constraints of generation technologies and power grid operations, but rather use aggregate model plants, mostly at a zonal levels (state, regional, and national), offering long-term projections. Also, PCM structure and level of granularity do not allow them to answer questions relative to long-term investment in new generation or transmission capacity. Instead, they capture detailed grid operations and offer short-term solutions to production cost minimisation.

While long-term planning helps shape the grid architecture (generation fleet and transmission and distribution networks), daily grid operations, in return, help (re)adjust long-term planning. There is therefore a critical need to examine the detailed system response to the various suggested projections, as it helps attest to the viability and soundness of these projections. This need is even more relevant for power grids in developing nations and their low reliability. Because PCM and CEM differ in terms of time scale and level of granularity, it is virtually impossible to verify long-term projections applicability in the short-term. Though they could be used in tandem, this procedure may require several iterations to arrive to optimal solutions (Mai et al., 2013), and considerable computational power, which may be impossible when dealing with large-scale power grids and longer planning horizons.

Diakov et al. (2015) present a tool attempting to link CEM and PCM and address their differences in granularity level. More specifically, the authors focus on taking ReEDS output and translating it into an input database for the PLEXOS production cost model. According to the authors, this tool's main goal is to impose regional generation projections outputted by ReEDS onto detailed, component-by-component, PLEXOS. Despite their efforts, the need is not quite addressed, given the differences in physical and temporal levels of details. Generation expansion projections are not always applicable, plants characteristics details not always compatible and transmission expansion are completely impossible to implement (Diakov et al., 2015).

Few other models seem to address the issue. The integrated planning model (IPM) developed by ICF, Inc. (EPA, 2018) is a long-term capacity expansion and production costing model used for analysing the North American electric power sector. It is a multi-regional, deterministic, dynamic linear programming model that performs system dispatch and operations simulation. Though relevant, the model does not capture the unit commitment problem. In addition, IPM:

- 1 assumes perfect competition
- 2 solves the economic dispatch problem with perfect foresight.

Assumption (1) means that the model overlooks any market imperfections such as market power. This is problematic at two levels. First, in a competitive market, sellers look to maximise their profits, and therefore based on their generation technology, may increase their influence in the market. Second, this assumption is based on the premise that all sellers/buyers have full knowledge about each other bid/offer, which is unrealistic. Assumption (2) implies perfect foreknowledge of future electricity demand, fuel supplies, and other variables, which, in reality, are subject to uncertainty. Hart and Jacobson (2011) propose a model to analyse the carbon emissions associated with high penetrations

of renewable energy, in the long-term. Their model uses deterministic planning, assuming that load can be predicted with no error, guaranteeing a power balance that ensures that the generation equals the load at all time. In addition, the model aggregates all available solar and wind data by location and assumes the same potential in all locations. In their study, Jacobson et al. (2015) support that the power system can be balanced with 100% penetration of renewables, with only wind, water, and solar resources, by year 2050. Their new grid integration model, LOADMATCH, offers low-cost, with no-load-loss solutions to this problem. However, LOADMATCH is not spatially explicit, as it does not capture the transmission network, and assumes perfect renewables forecasting. Also, it does not model operational reserves, which are essential in achieving reliability, especially when dealing with intermittent renewable sources.

It is safe to say that the complexity of power grid systems does not help in modelling such systems. While significant progress has been made in the recent years, further advances in grid modelling are needed, given physical and technological changes imposed by especially high penetrations of renewables, to warrant informed decisions. What is needed is a model, detailed enough to capture key day-to-day grid operations, yet abstract enough to enable fast execution. Such a model would do away with the difference in granularity level between CEM and CPM, addressing the issue of feasibility of power system operation in the short-term depending on the soundness of long-term expansion planning. *Spark!* offers that. It is a bottom-up discrete-event dynamic system model to simulate the behaviour of large-scale power systems, accounting for the variability of renewable energy resources, the thermal constraints of conventional generation resources, geographical information, and transmission network (Toba and Seck, 2018). The model also performs:

- 1 security-constrained unit commitment with a flexible look-ahead period, using an enhanced priority list method
- 2 economic dispatch.

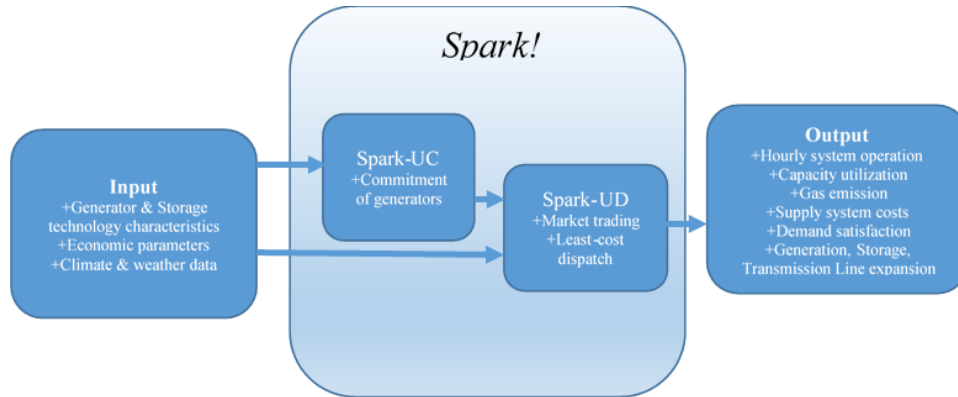
Table 1 shows differences between our model and the models mentioned.

2.2 *Spark!* model description

Spark! tackles two problems: the unit commitment on the one hand (tackled by the model component called Spark-UC), and the economic dispatch on the other (tackled by the model component called Spark-ED). The model is run in two modes, *forecast* and *actual*. In *forecast* mode, Spark-UC, based on forecasted meteorological (wind, water, solar) and load data, generating technology characteristics data, and transmission lines data, computes the *net load* and schedules the generating units needed to meet the remaining expected demand. The *net load* represents the system load not covered by renewables, rather by conventional sources. In mode *actual*, Spark-ED proceeds to dispatch power generators in real-time, based on operating costs and the availability of renewable energy resources, to meet current load. It dispatches generators with the lowest operating costs first, and the ones with the highest costs last, if needed. Supply and demand quantities are compared in order to assess an eventual need for power and engage in power trading, if

needed. This trading is carried out through bids to buy and/or offers to sell electricity if imports prove cheaper than use of local generators and storage units. Spark-ED executes multiple trading rounds until all offers are settled and no further trade is made. The system has reached equilibrium. Figure 1 shows the conceptualisation of these mechanisms.

Figure 1 Schematic of model mechanism (see online version for colours)



2.3 Model input

In this section, we define all input variable necessary for the model to run. Several database are used, where the model get information from. The data vary from season to season, and zone area to zone area.

- **Load forecast database:** power demand per consumer type, geographical zone and year. The different types considered here are residential, commercial and industrial sectors. This distinction is made given the different patterns of consumption. The residential sector refers to an energy-consuming sector that consists of private households. Commonly, we can think of space and water heating, air conditioning, lighting, refrigeration, cooking, etc. as main source of energy use. The commercial sector refers to consumption by non-manufacturing establishments or agencies mostly involved in the sale of goods or services (EIA, 2015). Example would be hotels, restaurants, schools, government buildings, etc. The industrial sector refers to consumption by manufacturing establishments.
- **Generator technology cost database:** expected fixed and variable operating and maintenance (O&M) costs (excluding fuel costs) for each generation technology. The fixed O&M costs are mainly planned maintenance and equipment costs, which vary with the plant size. The variable O&M costs are composed of outage maintenance (forced outage), consumables maintenance (equipment repair) and water supply costs (cost of providing cooling water for plant operations) (Rhyne, 2015). Start-up and shutdown costs are (costs associated with turning units on and off, respectively) also included in this database.

- *Generator technology database*: specifications of the technology used by the generators. Data include estimated lifetime, the heat rate, forced and planned outage rates, years of service, minimum uptime (minimum time the plant has to remain ON), minimum downtime (minimum time the plant has to remain OFF), ramp rate (time it takes for the plant to reach its nominal capacity) and minimal capacity (minimum capacity the generator must produce). We also consider *committable*, to specify whether or not a plant with a given generation technology can be committed. Here, renewable cannot be committed ahead of time as those technologies cannot be controlled. Technologies considered are, among others, wind, solar, coal, gas, geothermal, nuclear, biogas, biomass, oil and hydroelectric.
- *Fuel forecast database*: forecasted cost of all fuels. Fuels considered are oil, coal, gas, nuclear, biomass, biogas and petroleum.
- *Solar and wind potential database*: geocoded database that contains the potential for solar (exposure) and wind production (wind speed) for each geographical area covered by the model.
- *Transmission lines database*: all available high power lines. The information includes the remaining expected life of the lines, the losses, the associated costs as well as the lines capacity. The database indicates the origin and destinations of all the transmission lines.
- *Hydro flow database*: heads in dams and flows of water. Heads are change in elevation, between the upper and lower reservoirs.
- *Generator database*: all available generating units in the power grid network, as well as their characteristics. More specifically, we have the maximum capacity, name and technology used for power generation. The location for each generator is also indicated, as well as the year it was put in service.
- *Storage database*: this database is similar to the generator database, providing the same type of information, but in this case, for storage units.
- *Storage technology database*: data include storage efficiency (percentage of power which can be retrieved, out of what was stored) and discharge rate (rate at which the electricity is retrieved for usage), estimated years of service, forced and planned outage rates. Technologies include Mechanical (pumped-hydro, compressed air, etc.), electrical (capacitor, supercapacitor, etc.), electrochemical (flow battery, solid state battery, etc.), etc.
- *Storage technology cost database*: fixed and variable O&M costs for the storage technologies considered.

2.4 Model assumptions

Overall assumptions made in the model are as follow:

- We consider overnight costs, assuming the cost of a construction project incurs no interest, as if the project was completed ‘overnight’. This is a quick and simple way

to compare the costs of building different types of electrical plants, and ultimately, compute the total costs of operating them, after each run. This is a hypothetical scenario because a plant can obviously not be constructed in one night. We consider this option to evaluate the cost of a plant if it were built right away, with current prices, without taking into account the time it would actually take and how prices would rise over time.

- A reserve-margin constraint of 10% has been imposed in the grid. This choice is motivated by the study performed by the IRENA (2013b), for the year 2010, which considers the same value. Only conventional energy sources are considered to contribute to this requirement. This is because renewable sources cannot be committed and scheduled ahead of time.
- The load forecast error is assumed to be uniformly distributed. In choosing this distribution, we look to minimise chances to have a significantly important gap (difference between real load and forecast) at some time of the day, versus other time.
- Based on the input database, the model computes the limits of deployment and output characteristics of wind, solar, and hydropower technologies. Our model relies on mathematical models used in other peer reviewed studies to estimate the generation quantity. For wind technology for example, we resort to the Weibull probability density function (PDF), which has mostly been used to fit wind speed distributions for wind energy applications (Carrillo et al., 2014). Once the wind speed distribution is estimated, we also determine the power curve, which is done using a function defined in three regions (Cetinay et al., 2016). In order for the wind turbine to start generating power, wind speed must be greater than the cut-in speed v_{in} . Below this value, there is no power output. Similarly, the wind turbine stops operating at higher speed than the cut-off speed v_{off} . This is to prevent damage. Power is only generated in between v_{in} and v_{off} , with consideration of the rated speed v_{rated} , speed at which the maximum power output of the wind turbine is reached and generated before reaching v_{off} .
- The transmission system is modelled as a simple transportation model. Power shared between areas or zones in the network, via transmission lines, is only constrained by the capacity of those lines. This is in alignment with the level of abstraction of Spark!, which does not consider power flow principles and electrical components.
- The transmission loss is assumed constant throughout the scenario run. Normally, the losses vary, depending on specific conductors, the current flowing, and the length of the transmission line (Jackson et al., 2015). However, because our model has a higher level of abstraction, current and electric components are not captured.
- A power plant is equivalent to a generator, or a generating unit. This assumption is also justified by the model level of abstraction.
- Plants are assumed not to undergo maintenance or suffer breakdown. Forced and planned outages are not considered in the unit commitment and dispatch processes.

With the last assumption, we eliminate all errors in forecast and planning caused by planned and forced maintenance. Plants and transmissions are deemed in perfect operating condition at all time, during the whole scenario. The other assumptions are justified by the model level of abstraction and techniques used in peer-reviewed research addressing integrated resource planning issues.

3 Application

In this paper, we study the WAPP, which is composed of Economic Community of West African States (ECOWAS), namely Burkina Faso, Cote d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Niger, Nigeria, Senegal, Sierra Leone, Togo and Benin. Each of the countries in this grid would be considered as a zone. As any existing grid throughout the world, the main objective of the WAPP grid is to satisfy the populations' needs in each of these nations. The main question we attempt to answer in this research is the following: how to transform the West African power grid to meet the populations' economic, environmental, and security priorities, while making efficient use of the resources? These priorities are captured in the model outputs. Economic priorities are measured by *profits*, that is, financial advantage provided by operating the grid over a certain period of time. The higher the profits, the better. It indicates cost-efficiency. These profits are computed by the difference between total revenue of electricity sold to consumers and total spending, including fuel costs *fuelCost*, operation and maintenance costs *omCost*, fixed costs *fixCost* and overnight costs *ovrnCost*. Equation (1) displays the formula of the profit. Environmental priorities are measured by the amount of gas emissions *emission*, caused by electricity generation, mainly from fossil fuels. The less emission, the better, as it indicates less pollution. It is computed by the product of emission rate per power plants *eRate*, the quantity of power generated *utlzCap* and the heat rate *hRate*. Equation (2) displays the formula of the emissions. Security priorities are measured by the reliability of the grid, that is, the ability of the grid to meet the consumers' demands. For example, if there is a need for 10 MW and only 5 MW are met, the grid is said 50% reliable.

$$profit = \sum revenue - \sum ovrnCost - \sum fixCost - \sum fuelCost - \sum omCost \quad (1)$$

$$emission = \sum eRate \times utlzCap \times hRate \quad (2)$$

3.1 Model validation

Data used in this study are retrieved from the study conducted by IRENA (2013b). The model used was MESSAGE, to help investment alternatives via various scenarios, and assess economic, environmental and social implications of a given investment path. Scenarios were run from year 2010 to 2030. For validation, we use the method of model-to-model validation (Sargent, 2004). We compare our results with the ones obtained in the IRENA (2013b) study, at year 2010. Figure 2 displays the generation mix for year 2010 in the whole network, both from the WAPP study and *Spark!* Figures 3 and 4 display the generation mix for different countries, namely Senegal, Nigeria and Ghana. As shown in Figures 2–4, our model turns out to be very accurate, with very minor differences.

Figure 2 Generation mix for the year 2010 (see online version for colours)

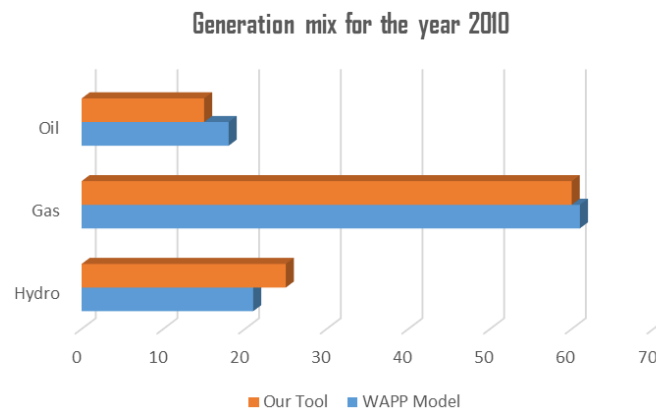


Figure 3 Generation mix 2010, by *Spark!* (see online version for colours)

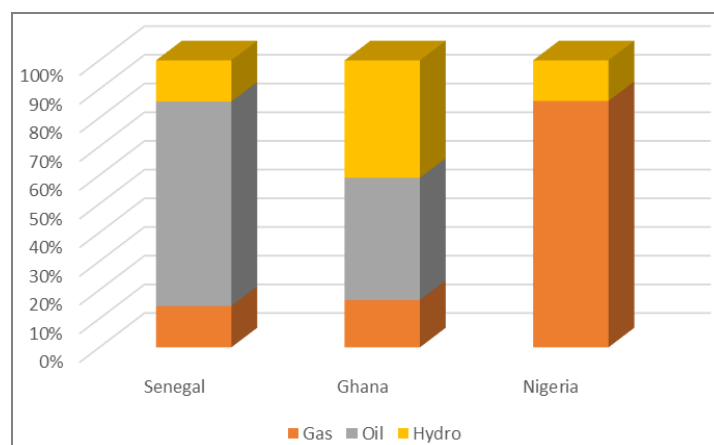
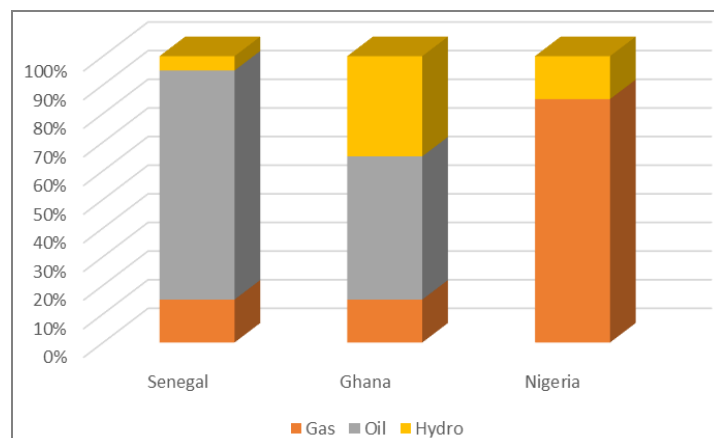


Figure 4 Generation mix 2010, by WAPP (see online version for colours)



3.2 Capturing grid architecture changes

Real world infrastructure systems are the product of an evolutionary process, highly interconnected and interdependent in nature, to a variety of factors, which make them far from optimal, but rather contingent on the system's history (Hathaway, 2000), as suggests the theory of path dependency. This theory supports that decisions faced today depend on historical choices made and past preferences, and are constrained by current knowledge (Mutizwa, 2017; Vergne and Durand, 2010). This theory attempts to explain how current situations are contingent on, and determined by past situations. Complex engineered systems, like power grid systems, come about as a result of historical, institutional and behavioural lock-ins and path-dependent technological development trajectories (Fouquet, 2016). Path dependency is, Simmie (2012) claims, what explains why it is proving so difficult to transition from fossil fuel-based energy systems to the general use of renewable energy.

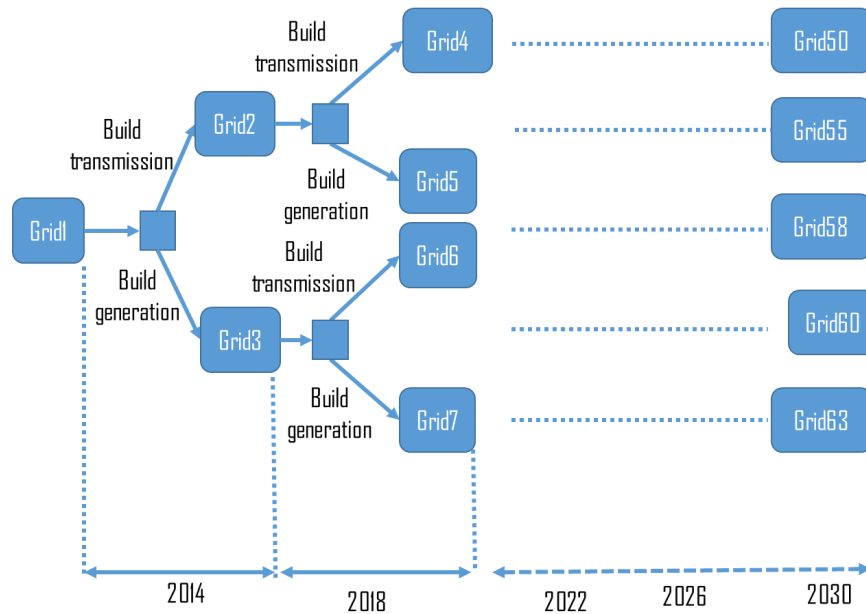
In this research, we propose a generative method that captures path dependency by iteratively creating grids, which are functionally identical yet structurally different, via decision making. The method would generate alternative grid architectures by subjecting an initial grid to a heuristic choice method for decision making, over a fixed time horizon. Grids are created, following decisions to either *build new generation capacity in zones* or *build new transmission lines between zones*. Each of this decision branches out, providing an alternative grid. Generation are added, based on resource potential in the zone, and are all renewables. Capacities added match the current deficit of electricity, while also taking into account the expected demand growth in the future. These grids are generated via a modelling framework. The model, *Spark!*, is used. The grids are generated in a 20-year long scenario (2010 through 2030), during which decisions to add capacities are made every four years. The aim is not to predict or replicate the actual evolution of the grid, but rather to create plausible grid architectures that can be compared in order to discover which types of expansion plans are more effective. In reality, decisions affecting the grid architecture are made more frequently, but using a smaller update period in the simulation would result in a number of alternatives that is too large to enable effective analysis.

At the beginning of the simulation (year 2010), the initial grid is run. Once run, power deficits are assessed, per zone, or country. If say, Ghana cannot cover its whole demand, decision is made to either build a new plant, with renewable resources – selection of the type of renewable is based on costs and availability – or build transmission lines coming from other countries to Ghana. In this case, not all countries are considered. Only the ones which are not in deficit, and not importing. Both alternatives are considered. One key information is that the same action is implemented in all zones. For example, if we follow option 'build generation', it is applied to all zones in deficit. Once all generation capacities are added in each corresponding zone, these zones altogether form a new grid. The grid operations are simulated. Then, deficits are again assessed. At that point, the two decisions are taken again, and so on, until year 2030. Figure 5 displays the tree-like exploration for grids generation. By year 2030, the grids generated are the results of transformations undergone in previous years, throughout the length of the scenario.

Scenarios used in other studies related to long-term planning in Africa (Bertheau et al., 2017; IRENA, 2013b; Ouedraogo, 2017; Panos et al., 2015, 2016; Taliotis et al., 2014) present different alternatives to be considered. Each of these alternatives are

mutually exclusive to the other. For instance, Panos et al. (2015) considers two scenarios, *Jazz*, with a focus on achieving economic growth through competitive and low-cost energy, and *Symphony*, with a focus on stronger policy regulations with priority given to sustainability and energy security. Both are run from year 2010 to 2050. These scenario differ in the changes of key variables (GDP, Per capita income, and CO₂ price) over time, with *Symphony* assumed to present more significant changes. These changes are applied to the whole network, for the whole scenario length. Unlike this type of scenario, our method enables decision making throughout the course of the scenario. Different set of actions can be taken, following a trajectory conditioned by previous decisions and current knowledge. We feel, it is a better representation of the impact of lock-ins and path dependency and a more accurate modelling abstraction of how grid system infrastructure come about and evolve over time.

Figure 5 Decision tree-like exploration (see online version for colours)



4 Modelling results

We look at the differences in performance between two generated grids, from the same initial grid in year 2010. We record the features of their architectures and analyse their associated performances. Let us call these grids gridX and gridY. Table 2 shows how both grids came about:

Figure 6 shows the energy mix in the network, at year 2010 and 2030, in gridX. Spark! displays the changes in electricity production shares over time. Though Figure 6 indicates an amelioration in renewable share in the overall network, this amelioration is not reflected in all nations, as displayed in Figure 7. Countries like Burkina Faso and Guinea are showing an increase in fossil fuelled generated power. In this graph, we can

identify nations that require attention regarding their energy mix, and warrant actions for remediation.

Table 2 Actions taken to generate gridX and gridY

Year	Actions	
	gridX	gridY
2010	Initial grid	
2014	Add generation capacity in all zones in deficit.	Add transmission capacity, aiming to all zones in excess, and which did not import power, to zones in deficit
2018	Add generation capacity in all zones in deficit.	Add transmission capacity, aiming to all zones in excess, and which did not import power, to zones in deficit
2022	Add transmission capacity, aiming to all zones in excess, and which did not import power, to zones in deficit	Add generation capacity in all zones in deficit.
2026	Add generation capacity in all zones in deficit.	Add transmission capacity, aiming to all zones in excess, and which did not import power, to zones in deficit
2030	Add generation capacity in all zones in deficit.	Add transmission capacity, aiming to all zones in excess, and which did not import power, to zones in deficit

Figure 6 Generation mix for the WAPP grid, (a) energy mix GridX – 2010 (b) energy mix GridX – 2030 (see online version for colours)

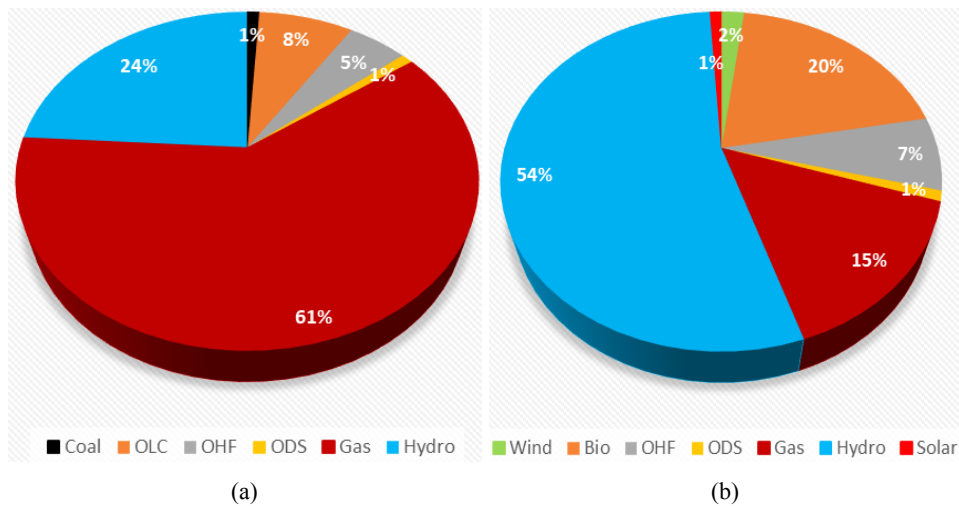


Figure 8 displays the level of reliability, in each zone. In this graph, we can compare how each country reliability level varies, between 2010 and 2030. Guinea, for instance, shows a drop in reliability, while Niger and Mali show improvement. Here too, we can identify local grids that need investments.

Figures 9 and 10 show the power generation in the Togo-Benin grid, illustrating how the grid responds to long-term projections made and how power plants are dispatched in real-time. Empty spots in the graph, between the demand line and imports indicate unmet demands. We can see that this local grid power dispatch has become more efficient in 2030 as more demands are satisfied.

Figure 7 GridX generation mix by country in 2010 and 2030 (see online version for colours)

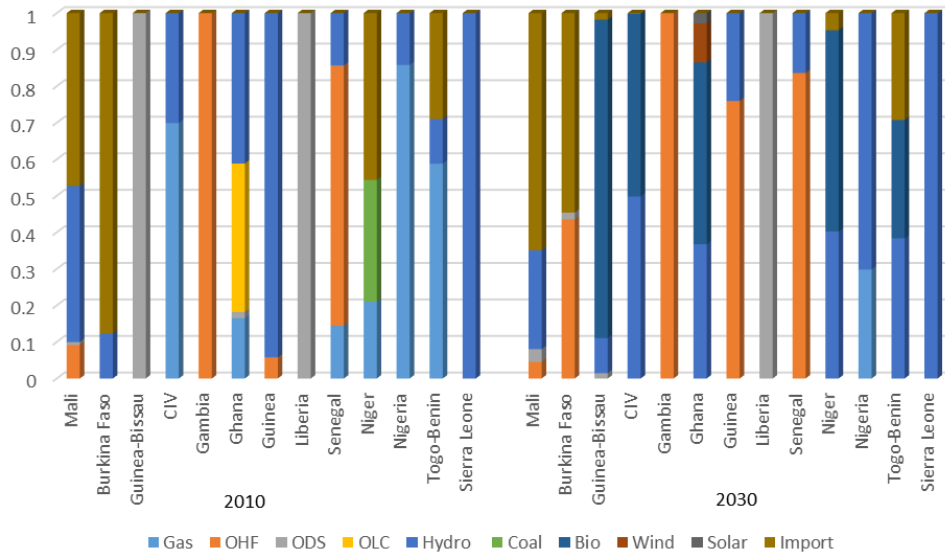


Figure 8 GridX reliability, per zone, 2010 and 2030 (see online version for colours)

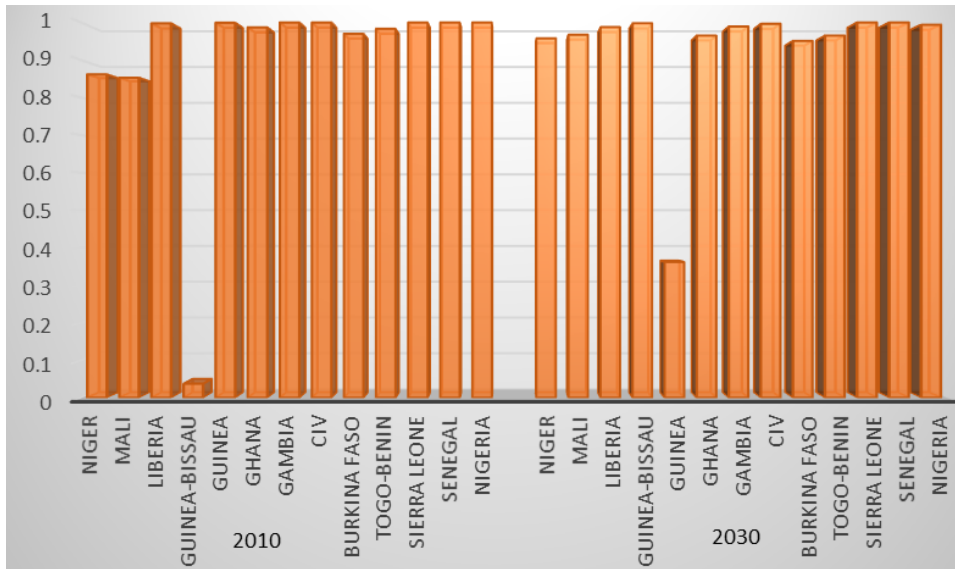
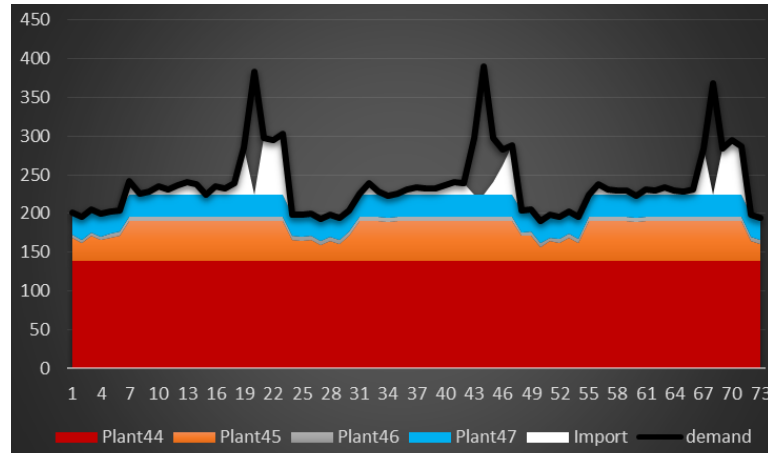
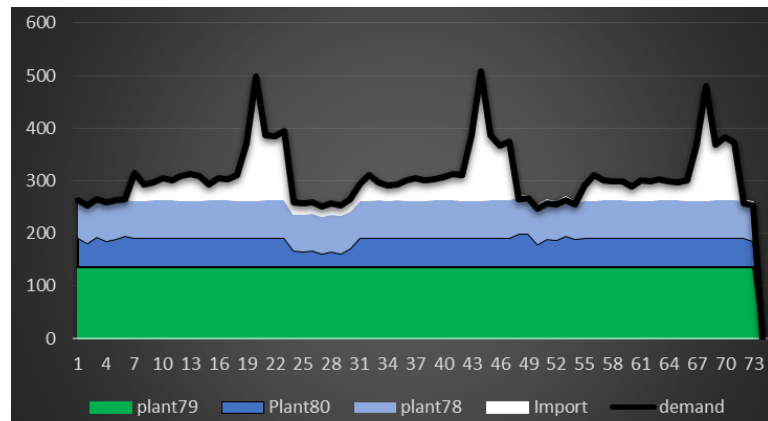


Figure 9 GridX power generation, Togo-Benin, 2010 (see online version for colours)**Figure 10** GridX power generation, Togo-Benin, 2030 (see online version for colours)

In Figure 9, plants 44, 45 and 46 are gas plants, while plant 47 is a hydroelectric plant. Gas plants are no longer operating in 2030, with instead, plants 79 which is biomass and plants 78 and 80, which are hydroelectric (Figure 10). We notice more imports and no more fossil fuelled generated power. This is an indication of an efficient grid transition, with good resource sharing and renewable penetration. Figures 11 and 12 display the gridX and GridY respective generation mix at different stages, up until in 2030. The model is able to show changes in resource use for electricity generation over time, in the whole grid. It helps track the level of renewable penetration, and also the level of use of fossil fuels. In Figure 11, we observe an increase in the use of gas and oil, and decrease in hydropower until year 2022. This is due to the fact that hydro plants were retired over these years, and available plants, oil and gas in this case, were dispatched to meet the demands. This changed by year 2026, with investments being made in hydro power, biomass, wind and solar, to replace retiring gas plants. Compared to GridX, GridY is obtained with more emphasis on transmission capacity addition. Figure 12 displays the generation mix of GridY, which indicates limited use of renewable sources, with an

increase use of oil over time. Because virtually no investment is in capacity generation, demands are met with only existing plant technology, which in this case, are mainly fossil fuel. Similarly to GridX, gas and oil usages are growing up until year 2022. After that, there is a drop in gas use, which remains constant until 2030. There is an increased use of oil over time.

Figure 11 Generation mix, GridX, year 2010 to 2030 (see online version for colours)

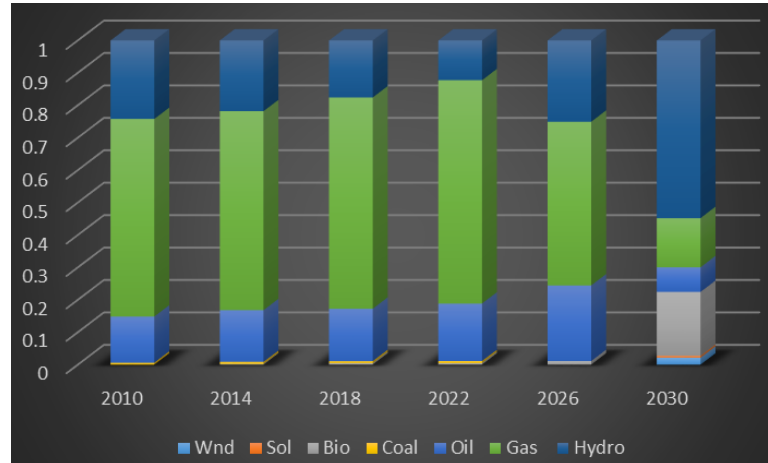
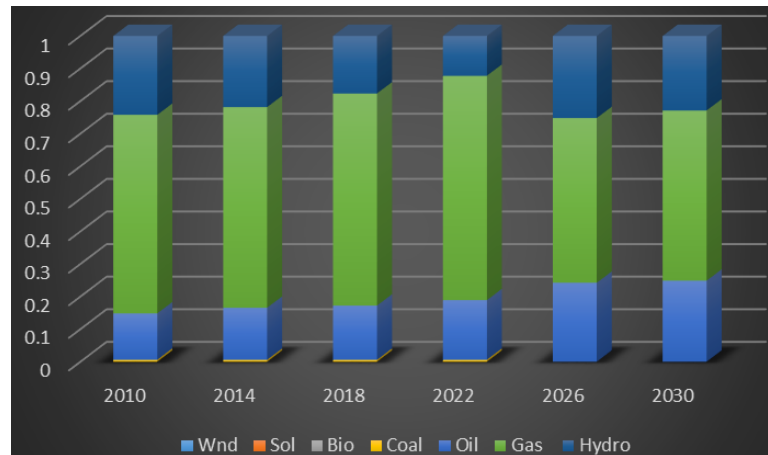


Figure 12 Generation mix, GridY, year 2010 to 2030 (see online version for colours)



In Figure 13 and 14, we observe the emissions of gridX and gridY, respectively, with a seemingly similar pattern through the scenario run. The period of 2010 to 2022 is marked by a rise in CO₂ emissions, given the mounting share of gas and oil to cope with the increase in demands. These emissions are reflected in the costs, with heavy fuel costs, limiting the profits. Year 2022 coincides with most of these plants being retired and in need of replacement. While renewable plants are added to the generation fleet in gridX, transmission lines are added in gridY. As a result, in both cases, emission levels drop

since in gridX, renewable do not emit, and in gridY, retired conventional plants are not replaced. Year 2026 and 2030 show improvement, with less fuel costs and emissions, due to hydro power usage.

Figures 15 and 16 display the financial performances of the grids, indicating more profits for GridX than GridY. These costs reflect the plants usage rate, and ultimately the energy mix. The higher the proportion of conventional source is in the energy mix, the higher the fuel costs, and consequently, the lower the economic gains in the network.

Figure 13 GridX gas emissions from 2010 to 2030 (see online version for colours)

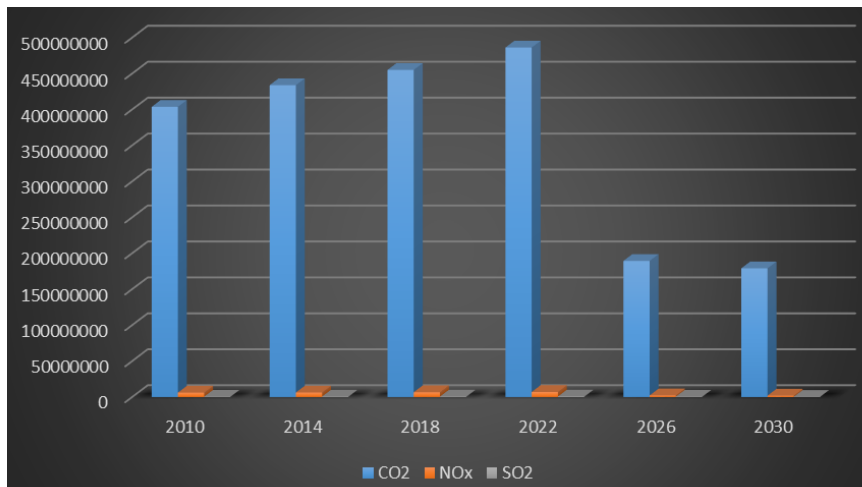


Figure 14 GridY gas emissions from 2010 to 2030 (see online version for colours)

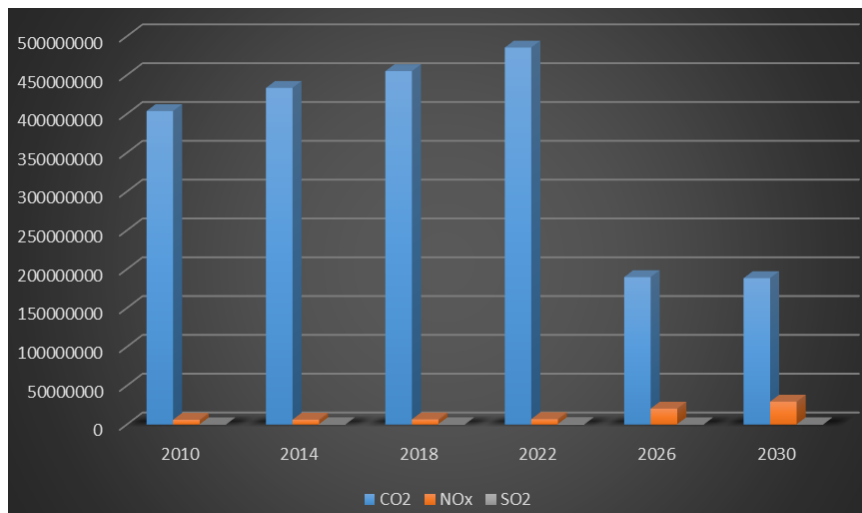


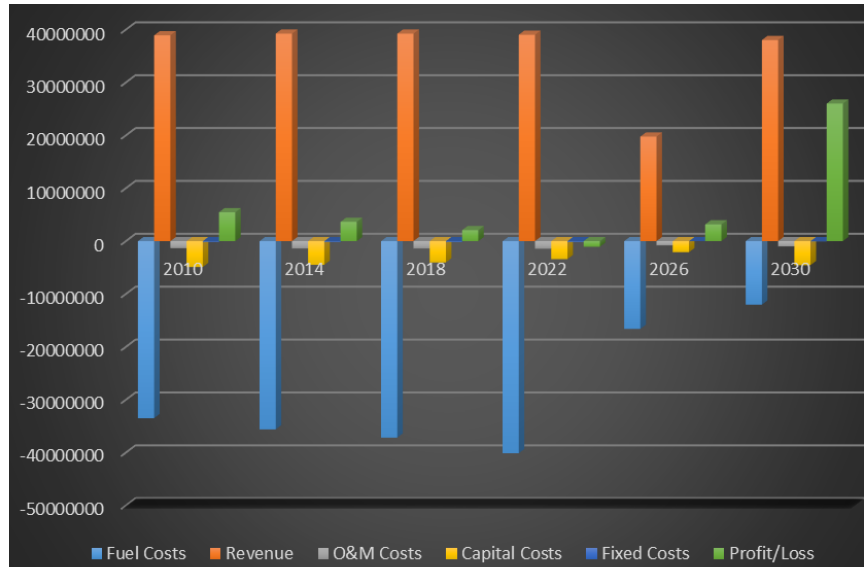
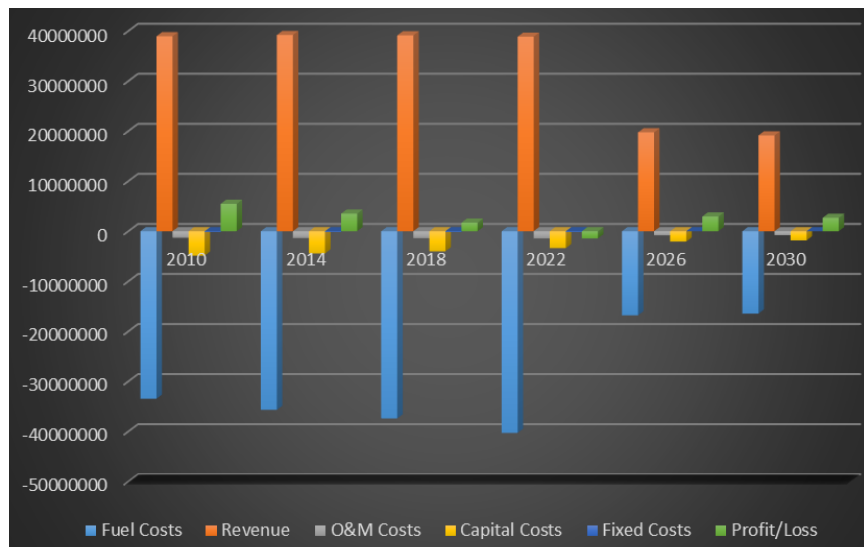
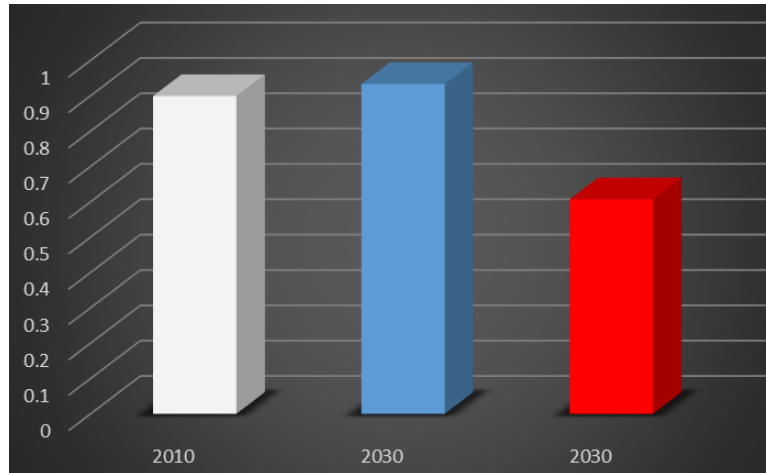
Figure 15 GridX financial performance from 2010 to 2030 (see online version for colours)**Figure 16** GridY financial performance from 2010 to 2030 (see online version for colours)

Figure 17 gives an overall view of the reliability, at the network level. There is an obvious improvement, which is a good indication of the efficiency of actions taken up to year 2030, for gridX. The drop in reliability for GridY is a result of the non-replacement of retired plants. Increasing demands are not being met.

Figure 17 Grid reliability, for the whole network, 2010 and 2030 (see online version for colours)

Note: Blue for GridX, red for GridY.

5 Conclusions

This paper develops a tree-like exploration method to analyse strategies for the development of grids of the future in West Africa. By adopting this approach, we look to identify alternatives to develop sustainable resource mix locally, and also lay the foundations for better interconnection across nations. Grids generated were run using *Spark!*, a power system simulation model for large-scale power system transition studies. The aim of this model is to offer a meso-scale solution that enables the study of very large power systems over long-time horizons, with a sufficient level of fidelity to perform day-to-day grid activities, and support architectural questions about the grids of the future. For WAPP grid, with no considerable investment in energy infrastructure, limited access to energy, underdeveloped power network, and inadequate generation systems, such a model presents great opportunities to address key issues.

Results show that under different course of actions, we will be faced with grids with different architectures and characteristics. Though GridX and GridY originate from a similar grid, different decision making took them on different trajectories, and to different destinations. GridX investments targeted mainly generation capacities, putting the focus therefore on local generation. In this grid, decisions enabled plants to be built anytime deficit was noticed, across the grid network, without too much regard for transmission. As a result, transition toward renewables could more easily happen. The integration rate of renewables were dependent on the occurrence of deficits, as anytime there is deficit, there is addition of renewable generation capacity. Not only this grid favours renewable integration, it also provides improvement in reliability level. This is not the case for GridY, where the focus was placed on interconnection. This grid has evolved to become much more interconnected, but with insufficient generation assets. Demands in the network could only be met by existing plants. However, when those happen to retire and

are not replaced, demands, which in the meantime are growing, remain unsatisfied. Such a grid not only limits renewable integration but also negatively affects electricity access.

Overall, policies for accelerated deployment of renewable energy are needed, not only to reduce costs, but also emissions. However, a careful selection of strategies that are appropriate to West Africa is also primordial as they are not equally sustainable. Our results suggest that long-term grid transition objectives are incompatible with transmission-focused decisions. It is in the global interest to increase generation using renewable sources, but also acknowledge the need for appropriate transmission network. Finding the right balance would be ideal, though complex. This study's main contribution lies there: Provide insight regarding decision trajectory which could lead to a stronger WAPP grid.

There are still room for improvement in this study. The topic of integrated resource planning is very important, especially given the emergence of renewable sources and issues related to their integration. One aspect to be looked at is forced and planned outage. It would be interesting to examine the effect of those factors on unit commitment and dispatch processes, and the repercussions on costs and reliability. This is particularly critical for grids in developing countries, given the level of technology and underdeveloped structure.

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